Proof Supplement: Provenance-Preserving Conservation Calculus and Non-Free Continuity

Whole-Event Inventories, Paid Syntax, Zero-Evaluation Certificate Inversion, and Step-Grouped Continuation

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Abstract

This proof supplement expands in full detail the technical constructions and step-by-step proofs invoked in Sections 3, 4, 5, 7, 8, 14, and 15 of *Provenance-Preserving Conservation Calculus and Non-Free Continuity: Whole-Event Ledgers, Observer Safety, and Well-Founded Settlement* (Second Edition, v2): Section A proves the four-way mathematical residual disposition induced by native-write receipts (WA12); Section B proves full-type restructuring certificate lifting, continuing transported remainders, and capacity-maintenance debit bounds (WA4–WA5, WA15–WA17); Section C proves owner-free syntactic birth, integer coefficient and substitution body replays, iterative batch charges, and all-count row authorities (WA13–WA14, WA18–WA21, WA26–WA28); Section D proves cross-language heterogeneous stock readback, same-receipt total writeback, exact completion distance and maximal future kernels, double-negation paid-AST recovery and primitive cochain replay (WA41–WA42), free constant-table integer certificate inversion (WA43), and contiguous step-interval anti-omission/anti-duplication batching with sealed-prefix tail continuation (WA44–WA45); Section E proves upstream resolvent-comb installation for long analytic windows (WA33). Shared process-core results cite Version 2 of the process-algebra companion [1] (DOI: 10.5281/zenodo.23297175).

A Four-Way Mathematical Residual Disposition Induced by Native-Write Receipts

A.1 Complete Linear Recovery and Three Exact Residual Witnesses (WA12)

Fix a concrete dynamical source D, state s, and target additive abelian group A. At every primitive event, the source input law supplies generator type G, the complete exposure tree rooted at that occurrence, initial seed, continuation tree, and evaluator tree; seed and continuation generate complete history H_s. None of these inputs contains a preset disposition branch, trivial-kernel premise, surjectivity premise, or external inverse.

Step 1 eliminates the whole-ledger compilation at that occurrence: because relation-write, continued-transport, redirect, and local-terminal payloads are all `PEmpty` in this vocabulary, the unique surviving native-write payload directly recovers $\varepsilon = D.\mathrm{emit}(s)$, and presentation round-

trip equality identifies the compiled receipt with the emitted event. Every algebraic object below is determined by that single transfer receipt.

Let $E : \mathbb{Z}^{(G)} \to A$ be the $\mathbb{Z}$-linear extension of the evaluator. History H_s retains both generator closure and relation closure: if the relation closure vanishes under E (sound relations), E descends to $\overline{E} : \widehat{H}_s \to A$ on the completed carrier; otherwise some actual relation q in the closure satisfies $E(q) \neq 0$. Faithful settlement classifies relation soundness, $\ker \overline{E}$, and $A/\operatorname{im} \overline{E}$ into four branches:

1. **Faithful isomorphism branch**: sound relations, $\ker \overline{E} = \{0\}$, and singleton cokernel $A/\operatorname{im} \overline{E}$ construct a full $\mathbb{Z}$-linear isomorphism $\phi : \widehat{H}_s \simeq A$ whose inverse satisfies $\overline{E}(\phi^{-1}(a)) = a$ for every $a \in A$.
2. **Unsound-relation branch**: extracts an actual relation q in the history's relation closure together with $u = E(q) \neq 0$.
3. **Nonzero-kernel branch**: sound relations with $\ker \overline{E} \neq \{0\}$ extract $k \in \widehat{H}_s$ satisfying $k \neq 0$ and $\overline{E}(k) = 0$ (zero numerical evaluation never erases this nonzero completed coordinate).
4. **Coverage-loss branch**: sound relations and $\ker \overline{E} = \{0\}$ with non-singleton $A/\operatorname{im} \overline{E}$ take the difference of two distinct cosets to obtain a nonzero coset $c \neq 0$ and preimage representative $a \in A$ ($[a] = c$), so $a \notin \operatorname{im} \overline{E}$. For **any** candidate recovery function $f : A \to \widehat{H}_s$ (without assuming linearity), $\overline{E}(f(a)) = a$ would imply $a \in \operatorname{im} \overline{E}$ and $[a] = 0 = c$, a contradiction; hence $\overline{E}(f(a)) \neq a$.

Because both material and residual projections install prior to the emitter on the same authoritative source, the ledger root and emitter D.emit remain unchanged, delivering alongside any branch the budget-preserving transfer:

$$\operatorname{entry}(s) \xrightarrow{\varepsilon} \operatorname{entry}(D.\operatorname{update}(\varepsilon)), \qquad \operatorname{next}(s) = D.\operatorname{update}(\varepsilon), \tag{S.1}$$

without mistaking residual classification for a strict same-debt debit.

B Full-Type Restructuring Certificate Lifting, Remainders, and Capacity Debits

B.1 Fieldwise Left-Inverse Lifting and All-Branch Whole-Join Folding (WA5, WA15–WA17)

In the finite complete-native append setting (WA15), augmenting row index set $I = \operatorname{Fin}(m)$ to $\operatorname{Fin}(m+1)$ preserves every prior row via $\operatorname{castSucc} : \operatorname{Fin}(m) \hookrightarrow \operatorname{Fin}(m+1)$ with its source/target entries, selectors, and receipts intact while placing the new math row at $\operatorname{last}(m)$. Repeated claims or identical payloads remain at distinct indices without deduplication; two-sided injectivity follows solely from the prior `IdentityScope` certificate and disjoint `old`/`math` tags.

In the general split/merge lifting setting (WA16), target injectivity is dropped. Augmenting mother vocabulary V to V^+ tags content, residual, bearer, beneficiary, scope, event, and lineage by `old` $\sqcup$ `math` while keeping every prior `Type`-valued admission, authority, assumption, origin, and progress, transfer, and discharge fibre verbatim on `old`. Embedding $j : \text{Obligation}_V \to \text{Obligation}_{V^+}$

and partial readback $p : \text{Obligation}_{V^+} \to \text{Option}(\text{Obligation}_V)$ satisfy fieldwise:

$$p(j(a)) = \text{some}(a), \tag{S.2}$$

proving j injective by applying p and cancelling `some`. Mother split child lists map to $\text{map}(j, \text{children})$ (and merge parent lists symmetrically to $\text{map}(j, \text{parents})$): list membership recovers mother membership via Equation (S.2), transporting source ownership, anchor transport, lineage, and local discharge directly; full origin-fibre coverage inverts j to invoke mother coverage (`math` targets cannot share an `old` origin tag); and no-phantom membership plus strict child budget descent $b(\text{child}) < b(\text{parent})$ commute with budget readout. The lifted restructuring certificate is thus generated entirely from the mother compiler image with zero target-side re-certification.

Across all continuing branches (WA17), a per-current `Packet` eliminates native-write, relation-write, continued-transport, or redirect from the generated ledger to recover (d, o), finite patch, and next current. Pairing math paid steps (or settled idle carries) with (d, o) commutes on prior entries:

$$d^+(j_s(e)) = j_t(d(e)), \qquad o^+(j_t(f)) = j_s(o(f)). \tag{S.3}$$

Listing only slot-0 math row explicitly and routing all prior rows to the compiled remainder event folds componentwise over the Σ-sum to Equation (S.3)'s exact whole join, lifting the mother restructuring certificate within the same step.

B.2 Capacity-Shortage Obstruction and Certified Maintenance-Debit Runs (WA4)

In `ResponsibilityCapacityKernel`, capacity readout $\mathcal{R}$ bundles source event, bearer, admitted obligations, demanded capacity $C_{\text{dem}} \in \mathbb{N}$, available capacity $C_{\text{avail}} \in \mathbb{N}$, and lists of incompatible demands, affected dependencies, and protected boundaries. Classifier $\mathsf{obstruction?}(\mathcal{R})$ returns $\text{some}\langle h \rangle$ if $h : C_{\text{avail}} < C_{\text{dem}}$ and none otherwise; decidability of $<$ on $\mathbb{N}$ proves both biconditionals in Equation (5) of the main text, while $\mathsf{CapacityObstruction}(\mathcal{R})$ holds only h with zero discharge or branch-selection fields.

For a carry-debit process B, each receipt $d : \mathsf{ActualMaintenanceDebit}(P, B, g, \lambda)$ holds slot $s \in \mathbb{N}$, live obligation o, source event ε, maintenance receipt r, `event_eq` identifying $g.\mathsf{lifecycleEvent} = \text{carried}(s, o, \varepsilon, \text{maintained}(r))$, pre/post slot liveness, $o.\mathsf{lineage} = \lambda$, and $B(\tau, \lambda) < B(\sigma, \lambda)$. Matching `event_eq` across `admit`, `reopen`, and the nine sub-cases of `cross` refutes eight branches by constructor clash and extracts `boundary_anchor_eq` plus `boundary_incidence_eq` in the sole surviving `maintained` branch to build `sourceOwnership`. Inductively mapping a certified run γ to $\mathsf{StrictBudgetRun}(B(\sigma, \lambda), B(\tau, \lambda))$ combines $B(\tau, \lambda) + 1 \le B(\mu, \lambda)$ with $|\text{prior}| + B(\mu, \lambda) \le B(\sigma, \lambda)$ to yield $(|\text{prior}| + 1) + B(\tau, \lambda) \le B(\sigma, \lambda)$, proving all bounds in Equation (6).

C Paid Syntactic Execution, Exact Node Charges, and All-Count Row Authorities

C.1 Telescoping Primitive Traces and Owner-Free Whole-Ledger Splicing (WA13–WA14, WA18)

Over multi-sorted abelian-group languages, remaining node counts satisfy:

$$r(\operatorname{const} c) = 0, \quad r(\operatorname{var} x) = 1, \quad r(e+f) = r(e) + r(f) + 1, \quad r(Le) = r(e) + 1, \quad r(B(e,f)) = r(e) + r(f) + 1. \tag{S.4}$$

Variable binds and constant binary, unary, and biadditive operations each consume one primitive step, while context steps recurse into left or right subtrees. Induction over step constructors proves $r(e) = r(e') + 1$ and $\mathrm{ev}_\rho(e) = \mathrm{ev}_\rho(e')$, so left-to-right evaluation produces $T_e : e \to \mathrm{const}(\mathrm{ev}_\rho\, e)$ with $|T_e| = r(e)$. Mapping $e \to e'$ to $\delta_e - \delta_{e'}$ telescopes along any prefix trace $T : e \to e_k$:

$$\partial\eta_T = \delta_e - \delta_{e_k}, \qquad \mathrm{ev}_\rho(\partial\eta_T) = 0. \tag{S.5}$$

Telescoping occurs solely in the formal boundary projection $\partial\eta_T$; every intermediate step, order, and relation index of T remains intact inside paid state (e_k, T_k). Extending support by optional math state gives budget $b_k = B - k > 0$ for $k < B := r(e)$ (emitting strict same-debt debits while prior rows carry via transported remainder) and local settlement at $k = B$ ($b_B = 0$, admitting no further paid macro-step). In the ten-node Fock instance at depth d, budget is $10 - (d+1)$ and live residual $\mathrm{physicalPair}_d - \mathrm{ev}_{\rho_d}(e_d)$ reduces at exhaustion to current minus origin physical pair.

C.2 Explicit Constants for Integer Expansions, Substitutions, Action Words, and Call Batches (WA19, WA21, WA26–WA28)

1. **Integer expansions**: $N_0(e) = \mathrm{const}\,0, N_{a+1}(e) = e + N_a(e)$ evaluates to $a\,\mathrm{ev}(e)$ at cost $a(r(e)+1)$; negative $z = -(a+1)$ adds one unary negation node, giving $c_z(e) = |z|(r(e)+1) + \mathbf{1}_{z<0}$. Folding support list $L = [(e_1, z_1), \ldots, (e_m, z_m)]$ into $F(L)$ yields:

$$r(F(L)) = \sum_{(e,z)\in L} (c_z(e) + 1), \qquad \mathrm{ev}(F(L)) = \sum_{(e,z)\in L} z\,\mathrm{ev}(e). \tag{S.6}$$

2. **Substitution body replay**: substitution θ at target environment Θ induces $\rho(x) := \mathrm{ev}_\Theta(\theta(x))$ and lifts every step in ρ to a trace in Θ—expanding each $\mathrm{bind}(x)$ into the **complete evaluation trace** of $\theta(x)$ so that $|T[\theta]| = r(e[\theta])$ at constant endpoints.
3. **Ordered action words**: for $|w| = n$, model-only ASTs use n letter nodes plus variable and projection $(n+2)$; prefix retention adds 1 $(n+3)$; paired trees pay $2n+5$. Letterwise effect history pays 5 on the empty word and 7 per prepended letter:

$$h([\,]) = 5, \qquad h(\ell u) = 7 + h(u), \qquad h(w) = 7n + 5. \tag{S.7}$$

4. **Cochain coordinates, batches, and child queries**: list $[\mathrm{updated}(e), \mathrm{old}(e)]$ ++ $\mathrm{mixedTerms}(e)$ injects mixed terms via $i \mapsto i+2$ at total vector cost equal to term costs

plus twice list length. Prefix pair-lifts $e_i = e[\sigma]^i$ with $\Sigma_n = \sum_{i=0}^{n} r(\tilde{e}_i)$ assemble into an $(n+1)$-coordinate vector batch of trace length $|T_{\text{batch}}| = \Sigma_n + 2(n+1)$, while $B+2$ actual calls pay their exact individual costs:

$$\text{callCharge} = \sum_{j=0}^{B+1} |T_j| = \sum_{j=0}^{B+1} r(\text{Raw}_j) = \text{callCost}. \tag{S.8}$$

Selected children of size m_i with output injection cost $4m_i+1$ (0 when absent), giving primary query cost $c_{\text{main}} = c_{\text{query}} + c_{\text{oldChild}} + c_{\text{nextChild}} + 3$.

C.3 Sealed Initial Row Authority and All-Count Recursive Admission (WA20, WA24)

Selecting slot 0 in the initial math patch directly certifies sealed initial authority a_0, and step $k \mapsto k+1$ advances (e_k, a_k) via canonical target entry and a_k.next without target-side re-admission. From settled material $M = (\rho, e, e', T, \text{owner}, \varepsilon)$, residual request $e + (-e')$ has positive budget $r(e) + r(e') + 2 \geq 2 > 0$, forcing a strict payment first step that compiles at that exact temporal occurrence to debtAdmission while preserving `firstDestination`, all prior projection outcomes, and canonical next (even when $\varepsilon = 0$, paying $B+2$ steps in full).

D Typed Stock, Maximal Future Kernels, Certificate Inversion, and Batch Continuation

D.1 Heterogeneous Readback of Cross-Language Stock and Born-Stock Offsets (WA22, WA25, WA32)

At every supplied occurrence, `stockLaw` outputs the complete dependent record holding upper/lower `Raw` inputs, both paid states and traces, variable bindings, migrated traces, and written stock. Inherited target outcome and source outcome satisfy `HEq`, yielding payload type equality; casting along that equality and eliminating `PEmpty` on the inactive branch composes with `HEq` to reduce to ordinary equality, recovering the exact cross-layer record. Under Equation (17)'s three nested seed seams, position j of the query trace lands at:

$$\iota(j) = j + |H_o| + |H_P| + 3, \qquad H_B[\iota(j)] = H_Q[j], \tag{S.9}$$

which is strictly monotone and injective by natural-number cancellation, while recursive folding $\text{historyWritten}(d+1) = \text{seed}(\text{historyWritten}(d), \text{written}(d+1))$ retains every historical stage $0 \leq d \leq \text{depth}$.

D.2 Same-Receipt Total-Value Writeback and Joint Difference Stock (WA23, WA29–WA31, WA34)

Paired evaluation outputs $(p, n) = (\mathrm{ev}_\rho\, e, \mathrm{eff}_{\rho,\varepsilon}\, e)$, where residual request $e - \mathrm{const}(p)$ settles at $\rho + \varepsilon$ to $n = \mathrm{ev}_{\rho+\varepsilon}\, e - p$, summing to the exact total:

$$p + n = \mathrm{ev}_{\rho+\varepsilon}\, e. \tag{S.10}$$

The decoder writes $(p + n)$ solely in the receipt-selected settled branch (retaining p when $n = 0$); the next born packet re-reads its own action result to refresh p^+ and its decoder, while `Move` aligns `target_root`, `whole_first`, prior projections, and `physicalNext` within one `generatedAction`. For two-execution boundary differences (WA30–WA31), $(r - c(\Psi, 0)) - (r - c(\Psi, \sigma)) = c(0, \sigma)$ executes its own integral-coefficient AST and embeds every new paid exposure through `pairWritten` and native stock into `jointStock`.

D.3 Exact Settlement Distance of `Type`-Valued Completion Receipts (WA32)

For any valid programme, frame, and start i with remaining pending node count B_i, strong recursion over `frame.action` returns $d = 0$ when settled and recurses on `mathNext` (budget $B_i - 1$) to return $d' + 1$ when paid, transporting endpoint settlement along $(i + 1) + d' = i + (d' + 1)$. Inducting over $0 \le h \le B_i$ gives $B_{i+h} = B_i - h$: positivity for $h < B_i$ excludes settlement so every intermediate step is paid, while $B_{i+d} = 0$ combined with $d \le B_i$ forces $d = B_i$.

D.4 Cofinal Stop Segment Coverage, Receiver Squares, and Maximal Future Kernels (WA35–WA37)

`StockTarget` compiles paid actions to `answered`/`mathNext` and settled actions to `debtAdmission`/`nextBorn`. Equation (20) ($t_0 = 0, t_{j+1} = t_j + 1 + d(t_j + 1)$) satisfies $t_{j+1} > t_j$ and $t_j \ge j$, covering every discrete step inside $[t_j, t_{j+1}]$ while incrementing `grade` solely at actual births. On $\mathcal{W}_j$, word action $S_j : \mathcal{W}_j \to \mathcal{W}_{j+1}$ reads the receiver's own environment $(\rho_j^{\mathrm{rec}}, \varepsilon_j^{\mathrm{rec}})$ via inverse transport:

$$\mathrm{receiverRead}_j(S_j w) = (\mathrm{ev}_{\rho_j^{\mathrm{rec}}}(w),\ \mathrm{eff}_{\rho_j^{\mathrm{rec}}, \varepsilon_j^{\mathrm{rec}}}(w)), \tag{S.11}$$

without assuming cross-grade environment equality. On $\mathcal{W} = \bigoplus_j \mathcal{W}_j$, $K_{\mathrm{fut}} = \bigcap_a \bigcap_j \ker(D_j S_a)$ is S-invariant, characterizes equivalence in $\mathcal{M}_{\mathrm{fut}} = \mathcal{W}/K_{\mathrm{fut}}$ by agreement at all (a, j), and embeds into the completion kernel at the empty word.

D.5 Three Environment and Positive-Credit Supplements in the Public Implementation

Three companion lemmas from the public v2 process-core implementation retain their exact byte identities in the whole-ledger package: (i) positive receiver credit follows directly from the two-node residual overhead (negation plus addition) minus the birth step; (ii) generic query/calculation environment alignment is stated explicitly and discharged automatically by `Incoming` in contextual families; and (iii) finite configurations with `calculationReader = none` wire directly to the incoming

calculation word of the same pair stock while keeping query readers, decoders, and both stocks intact.

D.6 Double-Negation Paid-AST Recovery and Primitive Cochain Differential Replay (WA41–WA42)

Cross-packet paid-method recovery and callback stock (`PaidMethod/{Source,Recovery,Admissions}.lean`, WA41). Given `baseFactory` and additive homomorphism $f : W_s \to_+ W_s$, define grade-n pair actions recursively by $\mathsf{actionAt}(f, 0) = f$ and $\mathsf{actionAt}(f, n+1) = \mathsf{pairAction}(\mathsf{actionAt}(f, n))$ ($\mathsf{pairAction}(g)(u, v) = (g(u), g(v))$), and define the three-branch syntactic extractor:

$$\mathsf{retainedCode}(n, \mathrm{frame}) := \begin{cases} \mathrm{linear}(-\mathrm{id}, u), & \text{if frame.registered.input.expression} = \mathrm{add}(\mathrm{add}(e_0, u), e_{\mathrm{eff}}), \\ \mathrm{linear}(-\mathrm{id}, u), & \text{else if frame.registered.input.expression} = \mathrm{add}(e_0, u), \\ \mathrm{frame.registered.input.expression}, & \text{otherwise.} \end{cases} \tag{S.12}$$

For any grade-n admission packet $P = \langle n, (s, \xi), \mathsf{cfg}, \langle \mathrm{rfl} \rangle \rangle$ settling at $p_\star = \mathrm{const}(v_{\mathrm{paid}}) \in V_{n+1}$, `nativeAfterAdmission` forms initial frame `native` with first-layer residual $\mathrm{add}(e_0, \mathrm{linear}(-\mathrm{id}, p_\star))$; with receipt distance d and settled frame selected $= \mathrm{frames}(\mathrm{native}, d)$, prefix invariance `paid_prefix_expression` proves selected keeps that first-layer residual, while `bornAfterNative` (P^+, at grade $n+1$) wraps a second outer residual $\mathrm{add}(\mathrm{add}(e_0, \mathrm{linear}(-\mathrm{id}, p_\star)), \mathrm{linear}(-\mathrm{id}, \mathrm{selected.event.state.1}))$. Applying the first branch of Equation (S.12) strips the outer residual and extracts $u = \mathrm{linear}(-\mathrm{id}, p_\star)$ to yield:

$$\mathsf{retainedCode}(P^+) \;\simeq\; \mathrm{linear}(-\mathrm{id},\, \mathrm{linear}(-\mathrm{id}, p_\star)), \qquad p_\star = \mathrm{const}(v_{\mathrm{paid}}), \tag{S.13}$$

which evaluates at any Γ^+ via $-(-v_{\mathrm{paid}}) = v_{\mathrm{paid}}$ (`negate_paid`) to v_{paid}. At grade $n+1$ (P^+), setting method input `raw f (n+1) frame` to $\mathrm{pairEnv}(\Gamma^+, 0)$ and $\mathrm{linear}(\mathsf{pairAction}(\mathsf{actionAt}(f, n+1)), \mathrm{liftExpr}(\mathsf{retainedCode}(n+1, \mathrm{frame})))$ uses $\mathrm{effect}(\Gamma, 0) = 0$ (`effect_zero`) and $f^{(n+1)}(0) = 0$ to output $(\mathsf{actionAt}(f, n+1)(v_{\mathrm{paid}}), 0) = ((f^{(n)}, f^{(n)})(v_{\mathrm{paid}}), 0)$ (`after_native_whole_action`). Setting `nextPairInventory` to `some (Stock.preserve prior (completeQueryWritten (n+1) frame))` embeds every event of `completeQueryWritten` into `birthStock` (`birth_keeps_callback`), while on contextual recognition families (`RecognitionAt H root`) sevenfold `InstallationAt` coface composition installs `InstalledFaces` prior to emission across every cofinal ordinal $k \in \mathbb{N}$ and finite segment `finite_paid_method_faces`.

Primitive-by-primitive word replay and cochain differential lockstep (`WordExecution.lean`, `RelationCochain/{Source,Consumer}.lean`, WA42). On any living source process with `readEnv`, observer environment $\rho_{\mathrm{obs}}(w)(t, x) := \mathrm{observer}(\lambda s.\, \mathrm{readEnv}(s, t, x), w)$ satisfies $(\mathrm{embed}(e)).\mathrm{eval}(\mathrm{env}(w)) = e.\mathrm{eval}(\rho_{\mathrm{obs}}(w))$ and $(\lambda t, x.\ \mathrm{binding}(t, x).\mathrm{eval}(\mathrm{env}(w))) = \mathrm{env}(\mathrm{sourceAction}(w))$. Recursing over all nine step constructors defines `embeddedStep` and `embeddedTrace`, which combine with `Trace.substitutedTrace` to produce `forward` and zero-increment pair lift `pairedForward` with exact length $|\mathrm{forward}(w, e)| = \mathrm{remaining}((\mathrm{embed}(e)).\mathrm{subst}(\mathrm{binding}))$ (`forward_charge`).

At frame `count` of the canonical particle–wave Fock sealed runtime `sourceRuntime`, inductive

clock alignment `actual_grounded` (starting from the initial 11-step calibration offset) identifies the active environment with step $11 + \text{count}$ of the underlying physical process, yielding `paired_binding_square`: $\text{sourceEnvironment}(\text{pairBinding}, \text{actualPairEnv}(c)) = \text{actualPairEnv}(c+1)$. Exposure `replayStock` enters both `birthStock` (`native_replay_written`) and the next complete query (`native_replay_next_query`), placing its boundary inside the next prefix history's relation closure as `actualBoundary`. Feeding $\mu_{\text{src}} = \text{sourceRelations}(c)$ through chain morphism τ_1 and invoking `SourceSubstitution.boundary_square` yields:

$$\text{relationMap}(\tau_1(\mu_{\text{src}})) = \text{substitution}(\text{relationMap}(\mu_{\text{src}})) = \text{actualBoundary}(c).\text{val} = (d^{0,1}(\text{actualBoundary}(c))).\text{val}, \tag{S.14}$$

delivering coimage differentials and `physicalReadback` at the same selected stop `stop = start + receipt.1`.

D.7 Free Constant-Table Relation Correction and Zero-Evaluation Certificate Inversion (WA43)

In `ConstantTransport.lean` and `ConstantInverse.lean`, let $G := \text{PairValue}(V, s) = V_s \times V_s$. In `ScalarRelationPresentation`, relation indices lie in $\text{RelationIndex}(\mathbb{Z}, G) := (G \times G) \sqcup (\mathbb{Z} \times G)$, with boundary $\text{relationMap} : \mathbb{Z}^{(\text{RelationIndex}(\mathbb{Z},G))} \to \mathbb{Z}^{(G)}$ defined on generators by:

$$\text{relationMap}([\text{inl}(u,v)]) = [u] + [v] - [u+v], \qquad \text{relationMap}([\text{inr}(z,g)]) = [z \cdot g] - z[g]. \tag{S.15}$$

Because $\text{freeEval} : \mathbb{Z}^{(G)} \to G$ sends $[g] \mapsto g$, both generators map to $0 \in G$, so $\text{im}(\text{relationMap}) \subseteq \ker(\text{freeEval})$.

1. **Forward free-table transport**: For (Γ, δ), define $\mathbb{Z}$-linear maps $F(\mathbb{Z}, G, X; s) \to \mathbb{Z}^{(G)}$ by $\text{valueMap}_\Gamma([e]) = [[\![e]\!]_\Gamma]$, $\text{effectValueMap}_{\Gamma,\delta}([e]) = [D_e(\Gamma, \delta)]$, and $\text{additionCertificate}_{\Gamma,\delta}([e]) = [\text{inl}([\![e]\!]_\Gamma, D_e(\Gamma, \delta))]$. On $[e]$, $[\![e]\!]_{\Gamma+\delta} = [\![e]\!]_\Gamma + D_e(\Gamma, \delta)$ gives:

$$[[\![e]\!]_\Gamma] + [D_e(\Gamma, \delta)] - [[\![e]\!]_{\Gamma+\delta}] = \text{relationMap}([\text{inl}([\![e]\!]_\Gamma, D_e(\Gamma, \delta))]); \tag{S.16}$$

   extending $\mathbb{Z}$-linearly and precomposing with coimage recovery $\text{recover} : \mathcal{W}_{\text{perf}} \to F(\mathbb{Z}, G, X; s)$ proves $\mathsf{computedAfter} := \mathsf{before} + \mathsf{effect} - \mathsf{corr} = \mathsf{after}$, matching `originFace.rootRead.source.newConstants` at $\widehat{w}$ via `F.whole_recovery` and preserving detector pairings (`actual_detector`).

2. **Explicit integer certificate construction and lossless inversion**: For any finite weighted list $L \in \text{List}(G \times \mathbb{Z})$ with word $W(L) := \sum_{(g,z)\in L} z[g] \in \mathbb{Z}^{(G)}$ and group sum $V(L) := \sum_{(g,z)\in L} z \cdot g \in G$, define `coefficientRelations(L)` recursively:

$$\begin{aligned} \mathsf{cert}([\,]) &:= -[\text{inr}(0,0)], \\ \mathsf{cert}((g,z) :: R) &:= -[\text{inr}(z,g)] + \mathsf{cert}(R) + [\text{inl}(z \cdot g,\, V(R))]. \end{aligned} \tag{S.17}$$

   Induction on L proves $\text{relationMap}(\mathsf{cert}(L)) = W(L) - [V(L)]$:

   - Base $L = [\,]$: $W([\,]) = 0, V([\,]) = 0$, while $\text{relationMap}(-[\text{inr}(0,0)]) = -([0 \cdot 0] - 0[0]) = -[0] = 0 - [0]$ (explaining why the empty list must retain $-[\text{inr}(0,0)]$ rather than 0: in the free module $\mathbb{Z}^{(G)}$, the basis vector $[0] \neq 0$!);

- Step $(g, z) :: R$: Equation (S.15) and the induction hypothesis expand the image to

$$-([z \cdot g] - z[g]) + (W(R) - [V(R)]) + ([z \cdot g] + [V(R)] - [z \cdot g + V(R)]) = (z[g] + W(R)) - [z \cdot g + V(R)],$$

cancelling $[z \cdot g]$ and $[V(R)]$ in $\mathbb{Z}^{(G)}$ to match $W((g, z) :: R) - [V((g, z) :: R)]$.

Enumerating the finite support of $\Delta T = \mathsf{computedAfter}(\widehat{w}) - \mathsf{before}(\widehat{w})$ in standard T1 as $L_{\Delta T}$ (with cert itself a pure list recursion requiring zero external kernel-preimage assumption) sets $\mathsf{actualCertificate} := \mathsf{cert}(L_{\Delta T})$ and $v_{\mathrm{paid}} := \mathrm{freeEval}(\Delta T)$, yielding Equation (90) ($\Delta T - [v_{\mathrm{paid}}] = \mathrm{relationMap}(\mathsf{actualCertificate})$). Eliminating the resulting `ExistsEvidence` witness `actualInverse` proves:

$$\mathsf{restoredTable} = \mathsf{before}(\widehat{w}) + [v_{\mathrm{paid}}] + \mathrm{relationMap}(\mathsf{actualCertificate}) = \mathsf{before}(\widehat{w}) + \Delta T = \mathsf{computedAfter}(\widehat{w}) = \mathsf{newConstants}, \quad \text{(S.18)}$$

with v_{paid} matching both the request's actual paid evaluation (`actual_paid_value`) and inverse-fibre value (`actual_inverse_value`).

D.8 Contiguous Step-Interval Anti-Drop/Anti-Dup Batching and Sealed-Prefix Tail Continuation (WA44–WA45)

Step-interval batch partition and anti-omission/anti-duplication gate (`Origin.lean`, `StepInterval.lean`, `StepInterval/Whole.lean`, `NextReceiptStepJointWhole.lean`, `Gate.lean`, WA44). Let $E = [e_0, \ldots, e_{m-1}]$ with start offset $s_0 \in \mathbb{N}$, item expansion $\mathrm{items}(e) \in \mathrm{List}(\mathrm{Expr} \times \mathbb{Z})$, and concatenated batch $R_{\mathrm{batch}} := \mathrm{requestsFrom}(s_0, E)$.

1. **Backward request origin (`readRequestOrigin`)**: Setting $P_k := \mathrm{requestsFrom}(s_0, E.\mathrm{take}\, k)$ and $Q_k := \mathrm{requestsFrom}(s_0 + k, (E[k]?).\mathrm{toList})$ gives $P_{k+1} = P_k +\!\!+ Q_k$ and $P_m = R_{\mathrm{batch}}$. For any $r \in \mathrm{Fin}(|R_{\mathrm{batch}}|)$, `OrderedTracePartition.locatePart` uniquely extracts block $b \in \mathrm{Fin}(m)$ and local index $\ell \in \mathrm{Fin}(|\mathrm{items}(e_b)|)$ satisfying:

$$r = |P_b| + \ell, \qquad R_{\mathrm{batch}}.\mathrm{take}\, r = P_b +\!\!+ ((\mathrm{items}(e_b).\mathrm{take}\, \ell).\,\mathrm{map}(\mathrm{form}_b)). \quad \text{(S.19)}$$

2. **Forward step intervals and two-way address identity (`stepInterval`, `sourceRequestAt_interval_address`)**: Splitting $E = (E.\mathrm{take}\, i) +\!\!+ [e_i] +\!\!+ (E.\mathrm{drop}\, (i+1))$ decomposes $R_{\mathrm{batch}} = \mathrm{prior}_i +\!\!+ \mathrm{own}_i +\!\!+ \mathrm{later}_i$. Setting $\mathrm{first}_i := |\mathrm{prior}_i|$ and $\mathrm{count}_i := |\mathrm{own}_i| = |\mathrm{sourceItems}(w_i)|$ isolates $G_i = \mathrm{own}_i$ with decoded request-word sum equal to $\mathrm{affine}_{\rho_{\mathrm{dec}}}(w_i)$, and cancelling prefix lengths in Equation (S.19) proves $r = \mathrm{first}_{i(r)} + s(r)$.
3. **Whole-batch folding and nonempty-group negative controls**: For every $(\mathcal{F}, P, \mathcal{E})$, event-list induction proves $\mathrm{flatten}([G_0, \ldots, G_{m-1}]) = R_{\mathrm{batch}}$ (`grouped_requests`, retaining empty $G_i = [\,]$ in place) and `executedTrace_grouped` gives $\mathrm{flatten}(\mathrm{map}(\mathrm{exec}, \mathcal{G})) = \mathrm{exec}(R_{\mathrm{batch}})$ (`grouped_executed_trace`); on the resolvent–price method instance over Section 14.1's conditional right-half-plane input domain (subject to Section 14.1's vacuous-truth boundary), finite-sum linearity in the Γ-successor space yields $\sum_{i=0}^{m-1} \mathsf{stepCochain}(i) = \partial_{\mathrm{joint}}$ (`actual_grouped_joint_boundary`). Any actual request $r \in \mathrm{Fin}(|R_{\mathrm{batch}}|)$ satisfies $0 \le s(r) < \mathrm{count}_{i(r)} = |G_{i(r)}|$, forcing $|\mathrm{own}| = |\mathrm{exec}(G_{i(r)})| > 0$; writing trace = before +\!\!+ own +\!\!+ after, strict length

inequalities $|\text{before}| + |\text{after}| < |\text{trace}| < |\text{before}| + 2|\text{own}| + |\text{after}|$ refute both omission (before ++ after) and duplication (before ++ own ++ own ++ after), closing `Gate.lean::actual_nonempty_group_rejects_omission_and_duplication`.

Sealed-prefix horizon extension and single-pass timeline (`NativeHistoryContinuation.lean`, `StoredHistory.lean`, `StoredTimeline.lean`, WA45).

1. Inductive type $\mathsf{HistoryAt}(F, s)$ (`exhausted` at 0, `step activated tail` at $F+1$) satisfies $h = \text{runFrom}(F, s)$ and `storedOccurrenceAt = occurrenceAt` by induction on F. `continueFor(h, E)` preserves all F `step` nodes verbatim and runs $\text{runFrom}(E, s_{\text{tail}})$ solely at `exhausted`; casting `continueFor(h, T-F)` along $(T-F)+F = T$ to define $\mathsf{extendTo}(h, T)$ proves:

$$(\mathsf{extendTo}(h, T)).\mathsf{prefix}(F) = (\text{runFrom}(T, s)).\mathsf{prefix}(F) = \text{runFrom}(F, s) = h. \tag{S.20}$$

2. For micro-step history $H : \mathsf{SourceNativeLivingRootHistoryAt}(F, s_0)$, `storedPairAt(H, i)` equals $(\text{advance}(i).\text{state}, \text{advance}(i+1).\text{state})$ (`stored_pair_actual`). Single-pass collector `collect` uses `Nat.dfoldRev` to push each state once into a capacity-$(F+1)$ array followed by the final state, forming $\mathcal{T} = \text{storedTimeline}(\mu, H) \in \text{Vector}(A, F+1)$ with $(\mathcal{T}[i], \mathcal{T}[i+1]) = (\mu(s_i), \mu(s_{i+1}))$ (`stored_timeline_adjacent`), whence applying $\mathcal{E}$ and `updateInventory` proves Equation (96) ($\mathsf{timelineInventory}(\mathcal{T}, i) = \mathsf{stageInventory}(\text{advance}(i))$).

E Upstream Resolvent-Comb Source Programme Interface for Long Analytic Windows

E.1 Closed-Facade Resolvent-Comb Installation and Trace Supply (WA33)

On the conditional right-half-plane nontrivial zero-observation input of Section 14.1, the nontrivial coordinate and right-half-plane fibre supply initial vector q_0, single-step resolvent-comb operator R, and exact iteration order $m \in \mathbb{N}$. In the single-variable environment binding $x \mapsto q_0$, recursive syntax $e_0 = \text{var}\, x, e_{k+1} = R(e_k)$ satisfies $\text{ev}(e_k) = R^k q_0$ and $r(e_k) = k+1$ by induction on k, so canonical evaluation at $k = m$ outputs target paid state, value $R^m q_0$, and a complete trace of length $m+1$.

Installing this calculation component prior to the emitter on `runtimeEffectAuthoritySource` lets the closed facade `RuntimeEffectFace` extract that exact result at the same occurrence (`runtimeCombCalculationReadoutAt_factorizes`), feeding target state, relation boundary, inverse fibres, and update cochains from that single $(m+1)$-step trace while preserving the lower physical ledger, emitter, and paired-Ω successor ($m+1$ measures syntactic node budget, while physical single-step transitions retain their own ledger receipts). Completed value $v = R^m q_0$ in turn generates the constant environment increment and residual request $R^m(v) - v$ feeding the window and inverse-residual chain of Section 14.

References

[1] Jian Gao. *The State Is Not the History, the Source Is: Identity, Responsibility, and Minimal Revision in Processes.* Version 2, 2026. DOI: 10.5281/zenodo.23297175. Main text and complete proof supplement.